INDUSENSE SYNTHETIC MAINTENANCE MANUAL
DOCUMENT ID: SYNTH-MAN-CNC-004
TITLE: Industrial Spindle Cooling Circuit & Thermal Dissipation Manual
STATUS: SYNTHETIC DATASET -- RESEARCH FRAMEWORK ONLY

SECTION 1: SYSTEM OVERVIEW
The 5-axis high-speed milling spindle employs a closed-loop ethylene glycol cooling circuit rated for continuous duty up to 24,000 RPM. Auxiliary heat exchangers dissipate stator and bearing thermal losses into the ambient bay atmosphere.

SECTION 4.2: HEAT DISSIPATION FAILURE (HDF) DIAGNOSTIC PROCEDURE
Trigger Criteria:
- Air Temperature: Ambient sensor T_air
- Process Temperature: Stator jacket sensor T_proc
- Rotational Speed: Spindle encoder N_rpm

CRITICAL ANOMALY RULE:
When the temperature difference [T_proc - T_air] falls below 8.6 Kelvin while the rotational speed remains below 1380 RPM, the system is in Heat Dissipation Failure (HDF) condition. Under low rotational velocity, reduced fan blade forced convection leads to thermal saturation.

MANDATORY CORRECTIVE ACTIONS:
1. Immediately disengage spindle drive and lock out main 400V breaker.
2. Inspect auxiliary cooling fan CF-02 for mechanical binding or blade debris accumulation.
3. Remove and wash suction filter mesh SF-12 using neutral solvent.
4. Verify glycol coolant line flow rate exceeds nominal 4.2 L/min threshold.
5. Record inspection sign-off before restarting spindle sequence.